

PROJECT PLAN

Project Title

An Intelligent NLP System for Legal and Employment Document Risk Analysis with Multiplatform Delivery

Phase I: Requirement Analysis and Literature Survey (Month 1)

The first phase focuses on understanding the project requirements and defining the overall scope of the system. A detailed literature survey will be conducted on Natural Language Processing (NLP), Large Language Models (LLMs), and existing LegalTech solutions. Research papers, journals, and recent technologies related to legal document analysis will be studied to identify research gaps. Functional and non-functional requirements will be gathered, and the Software Requirement Specification (SRS) document will be prepared. The overall system architecture, workflow, and technology stack will also be finalized. Relevant datasets consisting of legal and employment documents such as Terms of Service, Privacy Policies, Employment Agreements, and Non-Disclosure Agreements will be collected.

Deliverables:

- Literature Survey Report
- Software Requirement Specification (SRS)
- System Architecture Design
- Project Workflow
- Dataset Collection

Phase II: Data Collection and NLP Preprocessing (Month 2)

This phase involves developing the document acquisition and preprocessing modules. The system will support uploading legal documents in PDF, DOCX, and TXT formats as well as extracting content from website URLs. The extracted text will undergo preprocessing techniques such as text cleaning, tokenization, sentence segmentation, stop-word removal, lemmatization, and clause segmentation. The processed data will then be stored in the database to prepare it for AI-based analysis.

Deliverables:

- Document Upload Module
- PDF/DOCX/URL Text Extraction Module
- NLP Preprocessing Pipeline
- Cleaned Dataset
- Database Design

Phase III: AI Model Development and Risk Analysis (Month 3)

During this phase, Large Language Models such as Llama 3 or Mistral will be integrated into the system. AI models will analyze legal and employment documents to identify important clauses, including confidentiality, arbitration, indemnity, termination, intellectual property, liability, and non-compete clauses. The detected clauses will be categorized as Red Flags or Green Flags. The system will also generate simplified summaries, calculate Trust and Risk Scores, and provide recommendations to help users understand the document more effectively.

Deliverables:

- AI Analysis Engine
- Clause Classification Module
- Risk Scoring System
- Trust Score Generator
- Recommendation Engine
- Document Summarization Module

Phase IV: Web Application Development (Month 4)

A responsive web application will be developed using modern web technologies. The application will include user registration, authentication, document upload, AI analysis dashboard, visualization of detected clauses, Trust Score, Risk Score, summaries, and recommendations. Users will also be able to download comprehensive analysis reports in PDF format and access their previous analysis history.

Deliverables:

- Responsive Web Application
- User Authentication Module
- AI Dashboard
- Analysis Report Generation
- User Interface and User Experience Design

Phase V: Browser Extension and Database Integration (Month 5)

A browser extension will be developed to enable real-time analysis of Terms of Service and Privacy Policy pages while users browse websites. The extension will automatically extract webpage content and send it to the AI engine for analysis. MongoDB or Firebase will be integrated to securely store user profiles, document history, and analysis reports. The application will be optimized for performance, security, and seamless communication between the browser extension and the web application.

Deliverables:

- Browser Extension
- Database Integration
- User Profile Management
- Analysis History Module
- Secure API Integration

Phase VI: Testing, Deployment, and Documentation (Month 6)

The final phase focuses on testing and deployment. Unit testing, integration testing, system testing, and user acceptance testing will be carried out to ensure the system functions correctly. The AI model's performance will be evaluated and optimized. The complete application will then be deployed on cloud platforms. Finally, the project documentation, user manual, presentation, and final project report will be prepared for evaluation and demonstration.

Deliverables:

- Fully Deployed System
- Testing and Validation Report
- User Manual
- Final Project Report
- Project Presentation
- Demonstration

Project Title:**An Intelligent NLP System for Legal and Employment Document Risk Analysis with Multiplatform Delivery**

Phase	Duration	Objectives	Expected Deliverables
Phase I - Requirement Analysis & Literature Survey	Month 1	Identify project requirements, study existing LegalTech solutions, collect research papers, define system architecture, and gather legal & employment document datasets.	Software Requirement Specification (SRS), Literature Survey, Project Design, Dataset Collection
Phase II - Data Collection & NLP Preprocessing	Month 2	Develop document upload module, implement PDF/DOCX/URL text extraction, perform text cleaning, preprocessing, tokenization, and clause segmentation.	Text Extraction Module, Cleaned Dataset, NLP Preprocessing Pipeline
Phase III - AI Model Development & Risk Analysis	Month 3	Integrate Large Language Models (Llama 3/Mistral), classify legal clauses into Red Flags and Green Flags, generate summaries, recommendations, and calculate Trust/Risk Scores.	AI Analysis Engine, Clause Classification Model, Risk Scoring Module, Recommendation Engine
Phase IV - Web Application Development	Month 4	Develop responsive web dashboard with authentication, document management, visualization of analysis results, and report generation.	Complete Web Application, User Dashboard, Report Generation Module
Phase V - Application Browser Extension & Database Integration	Month 5	Develop Chrome extension for real-time ToS analysis, integrate MongoDB/Firebase, implement analysis history and user profile management.	Application, Database Integration, Analysis History Module
Phase VI - Testing, Deployment & Documentation	Month 6	Perform functional, integration, and performance testing, optimize AI responses, deploy application, prepare final report and presentation.	Fully Deployed System, Test Report, User Manual, Project Report, Final Presentation